1) k - Thermal conductivity

- The ability of a material to transfer heat energy through conduction

$$k = \frac{W}{m \cdot K} = \frac{kg \cdot m}{s^3 \cdot K} \underset{way}{\overset{in\ book}{=}} \frac{m \cdot L}{t^3 \cdot T}$$

α - Thermal diffusivity

$$\alpha = \frac{k}{\rho \hat{C}_p}$$ → measure the rate of heat transfer inside a material

$$= \frac{m^2}{s} \underset{way}{\overset{in\ book}{=}} \frac{L^2}{t}$$

$\hat{C}_p$ - Heat Capacity

- The amount of heat require to raise the temperature of a object by 1°C or a unit chang in its temperature

$$C_p = \frac{J}{k \cdot kg} = \frac{m^2}{s^2 \cdot k} \underset{way}{\overset{in\ book}{=}} \frac{L^2}{t^2 \cdot T}$$

q - Heat flux

- $q = -k \nabla T$

- This is the heat flux that is proportional to the temperature gradient

$$q = \frac{M}{t^3}$$

e - Combine energy flux vector

$$e = \Big(\underbrace{\tfrac{1}{2}\rho v^2}_{kinetic} + \underbrace{\rho \hat{H}}_{energy}\Big)v + \underbrace{[\tau \cdot v]}_{work} + \underbrace{q}_{heat}$$

$$e = \frac{M}{t^3}$$

2) $k_{gas} < k_{Liquid} < k_{solid}$

3)

Newton	Fourier
$\tau_{xy} = -\mu \frac{dv}{dy}$	$q_y = -k \frac{dT}{dy}$
- Transport of momentum	- Transport of heat
- Involve a vector field	- Involve scalar field
- Tensor	- Simple 1D scalar only
- Doesn't apply for solids	- Any Liquid, gas, solid

8)

$C_p - C_v = R$ is only for gas

For Liquids use $C_p - C_v = \frac{\alpha^2 T}{\rho \beta_T}$

9)

$$V_z = \frac{(P_0 - P_L)R^2}{4\mu L}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

V_{zmax}

$$V_z = V_{zmax}\left(1 - \left(\frac{r}{R}\right)^2\right) \rightarrow V_{z,Ave} = \frac{1}{\pi R^2}\int_0^{2\pi}\int_0^R V_{zmax}\left(1 - \left(\frac{r}{R}\right)^2\right) r\,dr\,d\theta$$

$$ke: \left(\frac{1}{2}\rho v\right)v = \frac{1}{2}\rho v_z^3 \qquad V_{z,Ave} = \frac{V_{zmax}}{2} \rightarrow 2V_{z,Ave} = V_{z,max}$$

$$\Rightarrow KE = \int_0^{2\pi}\int_0^R \frac{1}{2}\rho\left(2V_{z,Ave}\left(1-\frac{r}{R}\right)^2\right)^3 r\,dr\,d\theta$$

$$\frac{1}{2}\rho\, 8V_{z,Ave}^3$$

$$4\rho V_{z,Ave}^3 \cdot \frac{\pi R^2}{4}$$

$$= \pi R^2 \rho V_{z,Ave}^3$$

Put as $V_{z,Ave}$ for easy to measure as we already know the flow profile

$$KE = \alpha\left(\frac{1}{2}\dot{m}\, v_{avg}^2\right)$$

$$\dot{m} = \int_0^{2\pi}\int_0^R \rho V_{avg}\, r\,dr\,d\theta$$

$$\int_0^{2\pi}\int_0^R \rho\frac{V_{z,max}}{2} r\,dr\,d\theta = \rho\frac{V_{z,max}}{2}\pi R^2$$

$$= \frac{\pi R^2 \rho V_{z,max}}{2}$$

$$\Rightarrow KE = \alpha\left(\frac{1}{2}\pi R^2 \rho \frac{V_{z\,max}^3}{2}\right)$$

$$= \alpha\left(\frac{1}{4}\pi R^2 \rho V_{z\,max}^3\right)$$

Laminar $\alpha = 2 \rightarrow KE = \frac{1}{2}\pi R^2 \rho V_{z,max}^3$

$$= \pi R^2 \rho V_{z,Ave}^3$$

Same